



Contents lists available at ScienceDirect

Trends in Analytical Chemistry

journal homepage: www.elsevier.com/locate/trac

Ultrafast 2D NMR on a benchtop spectrometer: Applications and perspectives

Boris Gouilleux ^a, Benoît Charrier ^a, Serge Akoka ^a, François-Xavier Felpin ^{a,b},
Mireia Rodriguez-Zubiri ^a, Patrick Giraudeau ^{a,b,*}

^a Chimie et Interdisciplinarité : Synthèse, Analyse, Modélisation (CEISAM), UMR 6230, Faculté des Sciences, BP 92208, 2 rue de la Houssinière, F-44322 Nantes Cedex 03, France

^b Institut Universitaire de France, 1 rue Descartes, 75005 Paris, France

ARTICLE INFO

Keywords:

Benchtop NMR
Multi-dimensional NMR
Ultrafast
low-field NMR
Real-time monitoring
Edible oil
Screening

ABSTRACT

Benchtop NMR spectrometers are associated with significant resolution losses, as peak overlaps become ubiquitous at low field. 2D spectroscopy offers an appealing solution to this issue. However 2D NMR is associated with long experimental times which are ill-suited for high-throughput applications such as real-time reaction monitoring or rapid screening. The first implementation of ultrafast (UF) 2D NMR on a benchtop spectrometer –including B0 gradients– was recently reported, making it possible to record 2D spectra in a single –or at most a few– scans. In the present review, we investigate the analytical performance of UF 2D NMR at low field (43 MHz) and its application potential in two complementary research fields: real-time reaction monitoring and rapid screening. UF 2D spectroscopy at low field appears to be a powerful complement to existing analytical methods, and paves the way towards a number of developments in the field of spatially-encoded NMR at low field.

© 2016 Elsevier B.V. All rights reserved.

Contents

1.	Introduction	2
2.	Principle of UF NMR	2
3.	Analytical performance	3
4.	Reaction monitoring	5
5.	Screening applications	7
6.	Conclusion	9
7.	Material and methods	9
7.1.	Spectrometers	9
7.1.1.	400 MHz spectrometer	9
7.1.2.	43 MHz benchtop spectrometer	9
7.2.	Analytical performance	9
7.2.1.	Model samples	9
7.2.2.	UF 2D NMR experiments	9
7.2.3.	Conventional (Conv) 2D NMR experiments	9
7.3.	Reaction monitoring	9
7.3.1.	Procedures for the heck-matsuda reaction	9
7.3.2.	UF COSY experiments	10
7.4.	Screening applications	10
7.4.1.	Oil samples	10
7.4.2.	UF COSY experiments	10

Abbreviations: 2D, Two-dimensional; COSY, Correlation Spectroscopy; EPSI, Echo-Planar Spectroscopic Imaging; FTIR, Fourier Transform InfraRed; GLY, Glycerol; HF, High Field; HPLC, High Performance Liquid Chromatography; LF, Low Field; LOD, Limit Of Detection; NMR, Nuclear Magnetic Resonance; PUFA, Poly-Unsaturated Fatty Acids; RSD, Relative Standard Deviation; UF, Ultrafast; UFA, Unsaturated Fatty Acids; SNR, Signal-to-Noise Ratio.

* Corresponding author. Tel.: +33(0)2 51 12 57 09; Fax: +33(0)2 51 12 57 12.

E-mail address: patrick.giraudeau@univ-nantes.fr (P. Giraudeau).

<http://dx.doi.org/10.1016/j.trac.2016.01.014>

0165-9936/© 2016 Elsevier B.V. All rights reserved.

7.5. Details of data processing for ultrafast data	10
Acknowledgements	10
References	10

1. Introduction

Nuclear magnetic resonance (NMR) spectroscopy is one of the most powerful analytical tools, with applications in a broad range of domains, from the elucidation of (bio)chemical structures, to pharmaceutical, biomedical or food sciences. In the pursuit of a better sensitivity and spectral resolution, modern instruments are based on stronger and stronger magnetic fields generated by huge superconducting magnets. The amazing performance reached by these high field (HF) spectrometers is therefore associated with an expensive equipment and with the need for cryogenic fluids, involving a high monetary and environmental cost. Moreover, the incompatibility of this technology with harsh environments requires dedicated NMR laboratories, as well as a specialized staff to operate the spectrometers. Taken together, these economic and practical drawbacks have often kept NMR spectroscopy away from production and industrial sites.

Recently a new generation of low-field (LF) spectrometers, more compact, relatively inexpensive and cryogen-free, has brought NMR spectroscopy as close as possible to the production settings such as synthetic chemistry laboratories. Nowadays these so-called “benchtop NMR” spectrometers relying on permanent magnets, deliver ^1H spectra with a sufficient quality to enable promising applications in different fields such as chemical reaction monitoring [1–5] or screening applications [6–9].

However, the use of a low magnetic field is associated with a drastically reduced frequency dispersion as the static magnetic field B_0 decreases, leading to highly overlapped resonances. In addition to this overlapping issue, strong couplings are commonly encountered in LF NMR since the values of the coupling constants become comparable to the chemical shift dispersion. As a consequence, the identification and quantification of chemical compounds is far more complicated, and the resolution losses emerge as a limiting factor to extend the scope of LF NMR. In this context, NMR methods bringing a spectral resolution enhancement are particularly relevant in the low-field framework. By spreading the peaks over a 2D map, two-dimensional (2D) experiments [10,11] provide this advantage while delivering invaluable chemical insights on the analyzed compounds. Several homo- and hetero-nuclear 2D pulse-sequences have recently been developed on some benchtop instruments to take advantage of these assets at low field [2].

Nonetheless, the advantages of 2D NMR are associated with an intrinsic long acquisition duration, arising from the need to sample the frequencies involved in the indirect dimension through numerous 1D experiments in a time-incremented fashion. This long duration makes conventional 2D NMR unsuitable to study samples whose composition evolves in the course of the 2D sampling [12]. Beyond the subsequent time constraints, these conventional 2D experiments are not well adapted to reaction monitoring, which is a fundamental application of benchtop NMR. In order to alleviate this time constraint, Ultrafast (UF) NMR, which yields a whole homo- or hetero-nuclear 2D spectrum within a single scan [13], appears to be a natural candidate. At high field, UF 2D NMR has been applied to a variety of situations where the experiment duration is critical, from the real-time monitoring of chemical or biochemical processes [14] to the coupling with other analytical techniques such as HPLC [15] or Dynamic Nuclear Polarization [16], but also to high-throughput quantitative metabolomics [17,18]. Hence the development of the UF methodology on a benchtop spectrometer could generate a relevant analytical tool for numerous low-

field applications. Although UF NMR is nowadays well established as a powerful analytical tool [19], its implementation at low field is not straightforward since it requires a specific hardware, i.e. a powerful gradient coil which is not yet routinely available on benchtop spectrometers. Very recently, we reported the first UF COSY NMR experiment on a low-field instrument [20]. The spectrometer used is a prototype version of the commercial 43 MHz Spinsolve (Magritek, Aachen, Germany), which includes a gradient coil along the B_0 -axis designed for pulsed-gradient spin echo diffusion measurements.

This article aims at emphasizing the promising potential of UF experiments at low-field, as well as discussing the perspectives that could arise from this new analytical tool. The paper focuses on the COSY experiment, which is the most widely applied UF pulse sequence, and also the only one reported at low field so far. The first part describes the principle and basics of UF NMR, in order to provide an overview of the key features and characteristics of this generic approach. We also report the first evaluation of the analytical performance of UF NMR at low field. We then focus on the potential applications of this methodology in two domains where benchtop spectrometers appear particularly promising: reaction monitoring and screening analysis.

2. Principle of UF NMR

The main feature of the UF methodology is the replacement of the t_1 -incremented period in the conventional 2D NMR scheme by a spatially-dependent evolution period. To this end, the sample is virtually divided in several slices, which undergo different evolution periods according to their positions, all within a single scan (Fig. 1).

This spatial encoding is commonly performed in a continuous fashion by applying a magnetic field gradient G_z together with a linear frequency-swept pulse (chirp pulse), so that the spins located at different z positions are excited at different times $t(z)$. A constant-time spatial encoding scheme is generally used [21]. After a 90° conventional excitation, two successive chirp pulses applied with bipolar gradients induce a linear dephasing ϕ proportional to both the Larmor frequency Ω_i of the spins and their position z in the sample: $\phi(z) = C \cdot \Omega_i \cdot z$, where C is an adjustable constant depending on the spatial-encoding pattern [21]. After a conventional mixing period, a whole 2D NMR spectrum can be retrieved from the encoded information if the spin evolution during the detection period t_2 is collected in a way that refocuses the linear dephasing obtained through the spatial encoding step. To this end, an acquisition gradient G_a is applied while the receiver is open, leading to a train of echoes in the k space, where k is a wave number depending on the acquisition gradient: $k = \gamma_a \int_0^t G_a(t') dt'$ (γ_a is the gyromagnetic

ratio of the detected nuclei). These echoes arise when the refocusing induced by the acquisition gradient exactly compensates the dephasing ϕ previously provided by the spatial encoding step, i.e. $k = -C \cdot \Omega_i$. This series of echoes resembles a 1D spectrum –though obtained without Fourier transformation– corresponding to the first dimension of the UF spectrum, often called “ultrafast dimension”. The second dimension is then obtained through the application of an Echo-Planar Spectroscopic Imaging (EPSI) block during the detection. In this scheme, a train of bipolar gradient pairs enables to map the spatial dimension k and the time domain t_2 along a

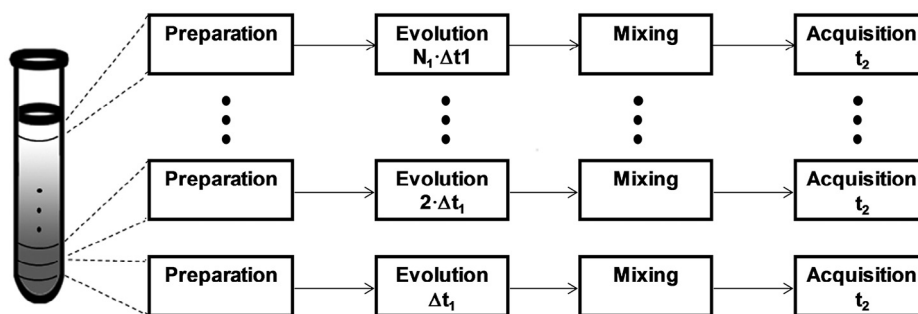


Fig. 1. Principle of UF 2D NMR. The experiment can be visualized as subdividing the sample into numerous slices that simultaneously undergo different experiments with an incremented t_1 delay.

zig-zag trajectory. This leads to a series of mirror-image 1D spectra with respect to the spatial dimension while the system evolves in t_2 under the influence of classical NMR parameters such as J-couplings, chemical shifts and transverse relaxation. After rearrangement of these series of sub-spectra, followed by a Fourier transformation, this t_2 evolution yields the second dimension (often called “conventional dimension”) so that a complete 2D spectrum is obtained in a single scan. For more details on the methodology and on the recent methodological developments which have significantly enhanced its performance, readers are referred to recent reviews on the subject [19,22].

By implementing this methodology with parameters properly adapted to a low-field spectrometer, we recently reported the first experiment based on an optimized constant-time phase modulated experiment at 43 MHz (Fig. 2) [20]. To illustrate this concept, we recorded a UF COSY spectrum in a single scan (0.4 s) on a sample of ethanol as a model compound. The quality of the spectrum shown in Fig. 2 highlights the suitability of the UF NMR experiments at low field, and the performance of the hardware. Since the values of coupling constants become non-negligible at low field compared to the frequency dispersion, the coupling patterns are visible along the conventional dimension, while they are usually not observed at higher field because of an insufficient resolution. The main drawback is the low sensitivity, however, this sensitivity loss is far less critical than what could be expected from the difference of one order of magnitude between magnetic fields. This is due in part to the probe design, but also to the fact that UF experiments are characterized by the need to compromise between resolution, sensitivity and spectral width [19]. In LF NMR, this compromise becomes more flexible as the spectral width is reduced. Moreover, the single-scan limit of detection is clearly high, but when the timescale of the targeted application is compatible with an acquisition duration from a few seconds to a few minutes, the single-scan LOD can be significantly increased by signal accumulation while avoiding t_1 -noise commonly encountered in the conventional 2D experiments. This hybrid approach, which has already shown its great interest at high field [23] should significantly enlarge the application perspectives of UF at low field. Note that it is still compatible with dynamic sample evolutions, since a complete 2D matrix is recorded at each scan –contrary to conventional 2D NMR– resulting in a final spectrum showing the average sample composition over the overall experiment time.

3. Analytical performance

This section aims at providing an evaluation of the analytical performance of UF NMR at low field, so that potential users can figure out what kind of applications can be expected from this methodology. As mentioned above, the high LOD of the single-scan acquisition is not compatible with most applications such as reac-

tion monitoring with realistic concentrations. Therefore, our analytical evaluation is based on UF COSY spectra recorded in 12 scans, leading to a total acquisition duration of 48 s. This duration remains very short compared to the typical duration of conventional 2D acquisitions and yields a LOD lower than 0.5 mol.L^{-1} that is more suitable for applications. The ester 3-bromo-ethylpropionate was chosen as a model compound, and five samples were prepared at different concentrations from 0.5 to 4 mol.L^{-1} in acetone- d_6 . Fig. 3 provides a qualitative comparison of the UF and

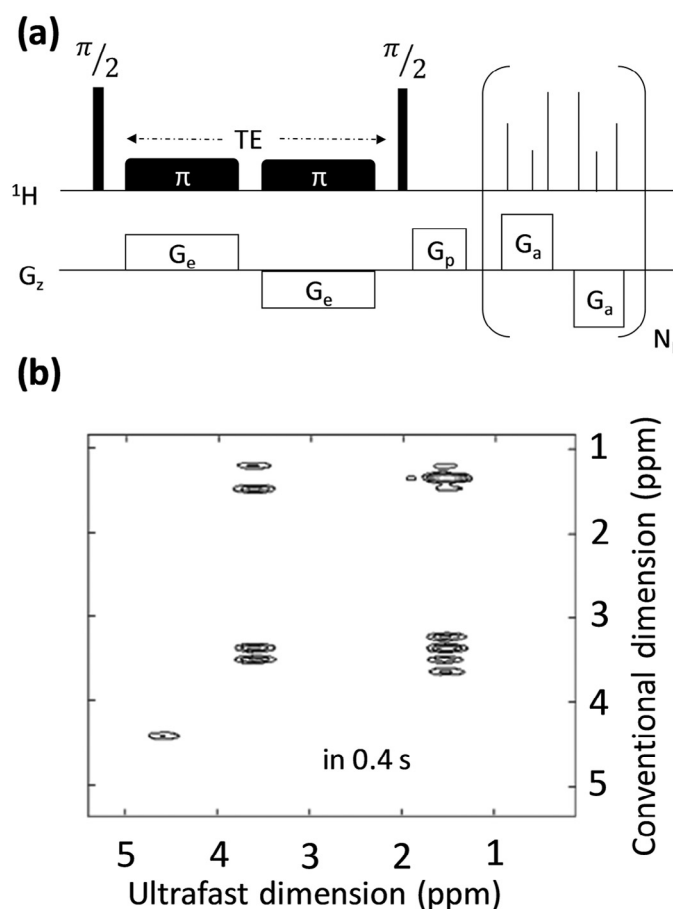


Fig. 2. UF 2D COSY pulse sequence (a) and corresponding single-scan spectrum (b) of a sample of ethanol in D_2O , recorded in 0.4 s on a 43 MHz benchtop spectrometer. The “ultrafast” axis refers to the spatially-encoded dimension (without Fourier transformation) whereas the “conventional” axis represents the direct dimension. TE: duration of spatial encoding; G_e : amplitude of spatial encoding gradients; G_p : amplitude of pre-acquisition gradient; G_a : amplitude of acquisition gradient; N_L : number of loops for the EPSI scheme applied during the signal detection.

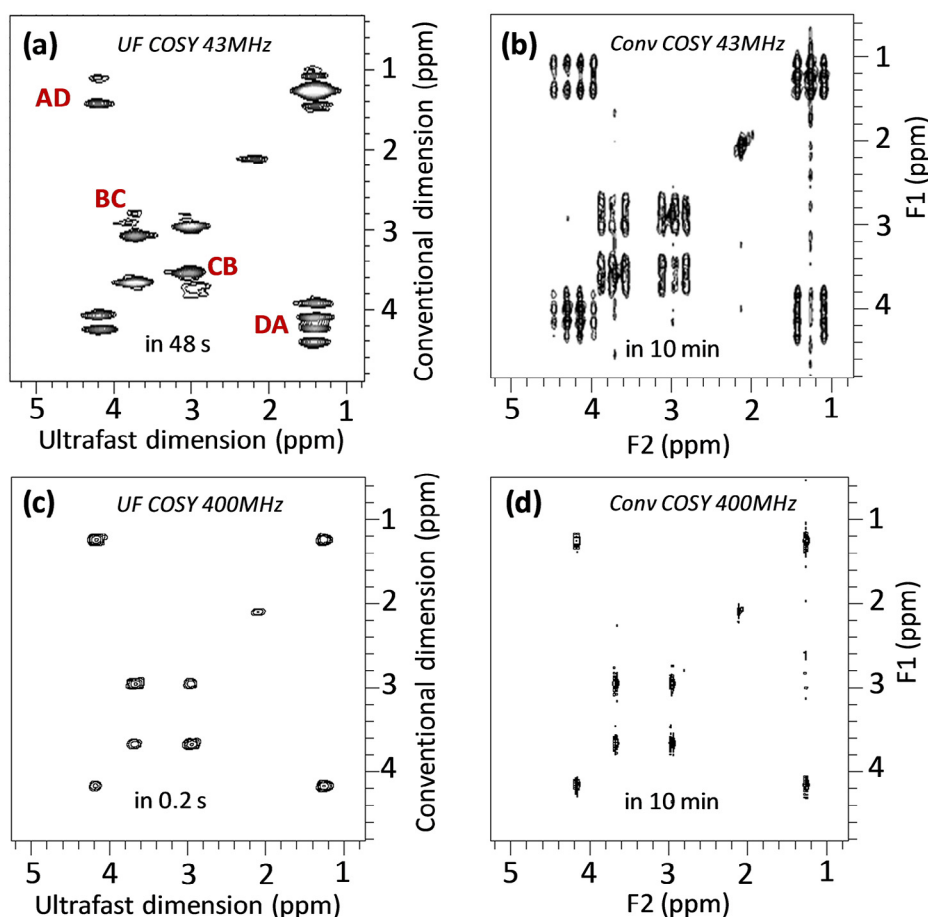


Fig. 3. 2D COSY spectra recorded on a 1 mol.L⁻¹ 3-bromo-ethylpropionate sample in acetone-*d*₆. (a) and (c) show the UF 2D spectra recorded with 12 and 1 scans at 43 MHz and 400 MHz, respectively, with the pulse sequence of Fig. 2a. (b) and (d) show conventional COSY spectra recorded in 10 minutes with conventional pulse sequences.

conventional COSY spectra recorded at 43 MHz (Fig. 3a and 3b) and at 400 MHz (Fig. 3c and 3d). The comparison between UF (Fig. 3a and 3c) and conventional spectra (Fig. 3b and 3d) at low and high field shows that field-induced resolution losses are far less important for UF COSY experiments than for its conventional counterpart. Indeed, at 43 MHz some of the 2D peaks are almost overlapped in the conventional spectrum while it is not the case for the UF COSY. This is mainly due to the fact that these UF experiments, relying on a “constant-time” encoding [21], are intrinsically decoupled along the ultrafast dimension [24], like any “singly pure-shift” experi-

ment [25]. Thanks to this feature, UF experiments provide an attractive way to discriminate the resonances at low field. To go further, this valuable methodology is now evaluated through quantitative analytical criteria.

The repeatability was evaluated for each sample by recording five experiments successively. Here the 2D peak volumes are considered without any normalization, so that the measured variations reflect all possible sources of irreproducibility. The results, presented in Table 1, are compared to those obtained with the same sample and in identical experiments at high field.

Table 1

Analytical performance of UF COSY experiments recorded in 12 scans (48 s) on 3-bromo-ethylpropionate samples in various concentrations and at two different fields (43 and 400 MHz). The repeatability is evaluated by the relative standard deviation (RSD) on the 2D peak volumes from five successive experiments. The linearity is determined after linear regression from the graph showing average 2D peak volumes as a function of the concentration. Peak labels correspond to the assignments indicated in Fig. 2

Cross-peaks			AD	DA	BC	CB
43 MHz	Repeatability (RSD)	4,0 M	1,19%	0,41%	0,33%	0,78%
		3,0 M	2,07%	0,25%	1,13%	0,34%
		2,0 M	3,95%	1,00%	2,81%	0,61%
		1,0 M	11,17%	2,57%	2,98%	2,67%
		0,5 M	6,06%	5,56%	6,10%	6,71%
	Linearity (R ²)		0.9714	0.9893	0.9823	0.9906
400 MHz	Repeatability (RSD)	4,0 M	7,74%	7,37%	8,07%	8,18%
		3,0 M	5,18%	4,73%	5,24%	5,31%
		2,0 M	2,12%	1,86%	2,15%	2,06%
		1,0 M	5,03%	4,12%	5,13%	5,34%
		0,5 M	3,34%	3,37%	3,51%	3,67%
	Linearity (R ²)		0.9911	0.9947	0.9921	0.9921

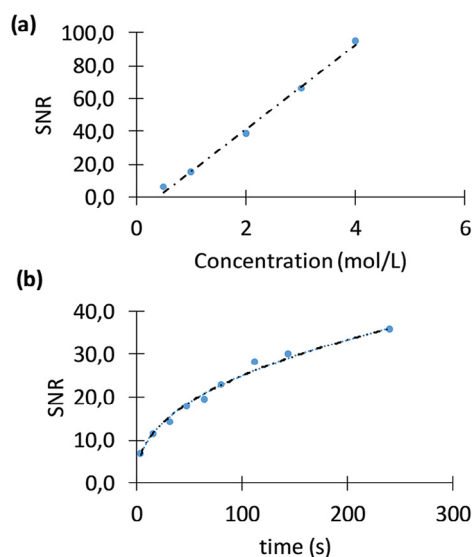


Fig. 4. Typical SNR determined from UF COSY spectra on model 3-bromoethylpropionate samples in acetone- d_6 . (a) SNR as a function of concentration for a 48 s experiment recorded with 12 scans. (b) SNR as a function of the experiment time measured on a 1 mol.L⁻¹ sample. The SNR was computed on the signal CB as follows: $S(\text{CB})/\sigma(\text{noise})$, where $S(\text{CB})$ is the amplitude of the peak CB and $\sigma(\text{noise})$ the standard deviation of the noise evaluated on an area of $(F1 \times F2) = (0.65 \text{ ppm} \times 1.14 \text{ ppm})$ centered at $(F1; F2) = (4.6 \text{ ppm}; 3.0 \text{ ppm})$.

At low field, the repeatability mainly depends on the concentration, i.e. on the signal-to-noise ratio (SNR). The RSD is in general higher when the concentration decreases. Yet, the repeatability does not exceed 3% for most of the peaks at 1 mol.L⁻¹. The situation is quite different at high field where the worst repeatability is obtained at the highest concentration. Here, the SNR is not a limiting factor for such highly concentrated samples, and the lower repeatability for the most concentrated samples is probably explained by radiation damping effects –which can be suppressed if needed by appropriate methods [26]. But on the whole, Table 1 results show that the repeatability of UF COSY at low field is comparable to the one measured at high field.

The method linearity was also evaluated from the 2D peak volumes measured on the five model samples. Five spectra were recorded, and for each concentration the average peak volume was calculated. The high coefficients of determination highlight the good linearity of the UF COSY both at high and low fields, with a slightly higher performance at high field.

Fig. 4 gives a more precise idea of the SNR values that can be measured as a function of concentration, but also as a function of the experiment time –directly related to the number of scans. As expected, the SNR evolves linearly with concentration, while evolving with the square root of the number of scans when signal averaging is carried out.

In conclusion, the analytical performance (in terms of repeatability and linearity) appears similar to that of identical experiments at high field. The limit of detection is of course higher for LF UF, but can be decreased at a reasonable time cost when several scans are recorded. These results open the way to promising analytical applications, while highlighting the concentration limits of the method.

4. Reaction monitoring

Reaction monitoring performed through analytical spectroscopies yields essential kinetic and speciation information which could be determinant in numerous applications such as organic and inorganic synthesis, and for chemical engineering [27]. The

development of accessible, powerful and robust analytical methods is therefore important to identify and quantify, in real time, the compounds involved in chemical processes. NMR spectroscopy has the advantage of being non-specific and non-destructive; moreover it requires no prior knowledge on the analyzed compounds. Furthermore, it delivers invaluable structural and quantitative information. HF NMR has been widely used in organic and pharmaceutical applications for monitoring chemical processes, either directly inside the NMR tube [28–30] or by using flow cells [31–35]. In spite of the valuable analytical insights, NMR remains less used than other more accessible spectroscopies such as FTIR or Raman, since it generally requires sophisticated and customized monitoring systems set up in dedicated laboratories.

It is against this background that benchtop NMR has emerged as a promising processing analytical tool; however its applicability is reduced by a lower performance compared to its HF counterpart. So far, resolution losses and overlapping issues have limited the application range of LF NMR to the monitoring of relatively simple reactions where the peaks of interest are well isolated, unless deconvolution tools are employed [3–5]. In this context, the potential of UF 2D NMR seems worth exploring.

We recently reported the suitability of this methodology through the monitoring of a Pd-catalyzed Heck-Matsuda reaction (Fig. 5a) performed in the NMR tube [36]. This variant of the Heck coupling uses aryl diazonium salts as highly reactive aryl halide surrogates, and provides reactions in mild conditions under ligand- and base-free conditions [37]. Moreover, this reaction has recently found a growing interest, especially since its hazardous character was alleviated by a bicatalytic strategy [38,39] and/or a continuous-flow approach [40,41].

In this reaction, the aniline **1** dissolved in methanol first reacts with *t*-BuONO and MeSO₃H to form a diazonium salt. The latter is then coupled with methyl acrylate **2** in the presence of Pd(OAc)₂ as a catalyst, in order to produce the styrene **3**. The experiment must be designed to keep the mixture as homogeneous as possible in order to avoid disruption of the NMR acquisition. It is important to notice that thanks to an external lock system, experiments are carried out in a non-deuterated solvent. This reduces the cost of the monitoring and allows to avoid unwanted potential isotopic effects that could affect reaction rates in deuterated solvents.

The Heck-Matsuda reaction represents a well-suited model reaction to assess the relevance of the UF approach at 43 MHz. Indeed, this reaction provides relatively crowded resonances in the aromatic region, justifying the need for multi-dimensional experiments. Moreover, the reaction conditions –use of a metallic catalyst and gas formed during the coupling reaction– represent a challenging example. Here, in view of the relatively low SNR obtained in a single scan, the experiment duration was adapted to the reaction timescale (36 scans with a recovery time of 1.25·T₁, leading to the acquisition of a COSY spectrum every 2.6 min). As previously reported, the sensitivity was optimized through the optimization of J-modulation effects [42]. The 2D UF spectra shown in Fig. 5 highlight how this methodology can deliver time-dependent and spectrally resolved cross peaks, both for the product **3** and for the limiting starting material **1** (Fig. 5b, 5c, 5d). The starting material in excess **2** is also visible on the spectra.

In this preliminary work, the reaction monitoring was performed directly inside an NMR tube under mild reaction conditions, i.e. 29°C, with an initial concentration of 0.36 mol.L⁻¹ for the limiting starting material. The kinetic curves recorded from the time-evolution of 2D peak volumes are shown in Fig. 5e. These curves depict the two chemical processes involved in the Heck-Matsuda reaction very well. The decreasing blue curve matches with the initial conversion of **1** into the corresponding diazonium salt. After the aniline is consumed (at ca. 20 min), the coupling reaction starts, leading to the final styrene **3** (increasing pink curve). The trend of

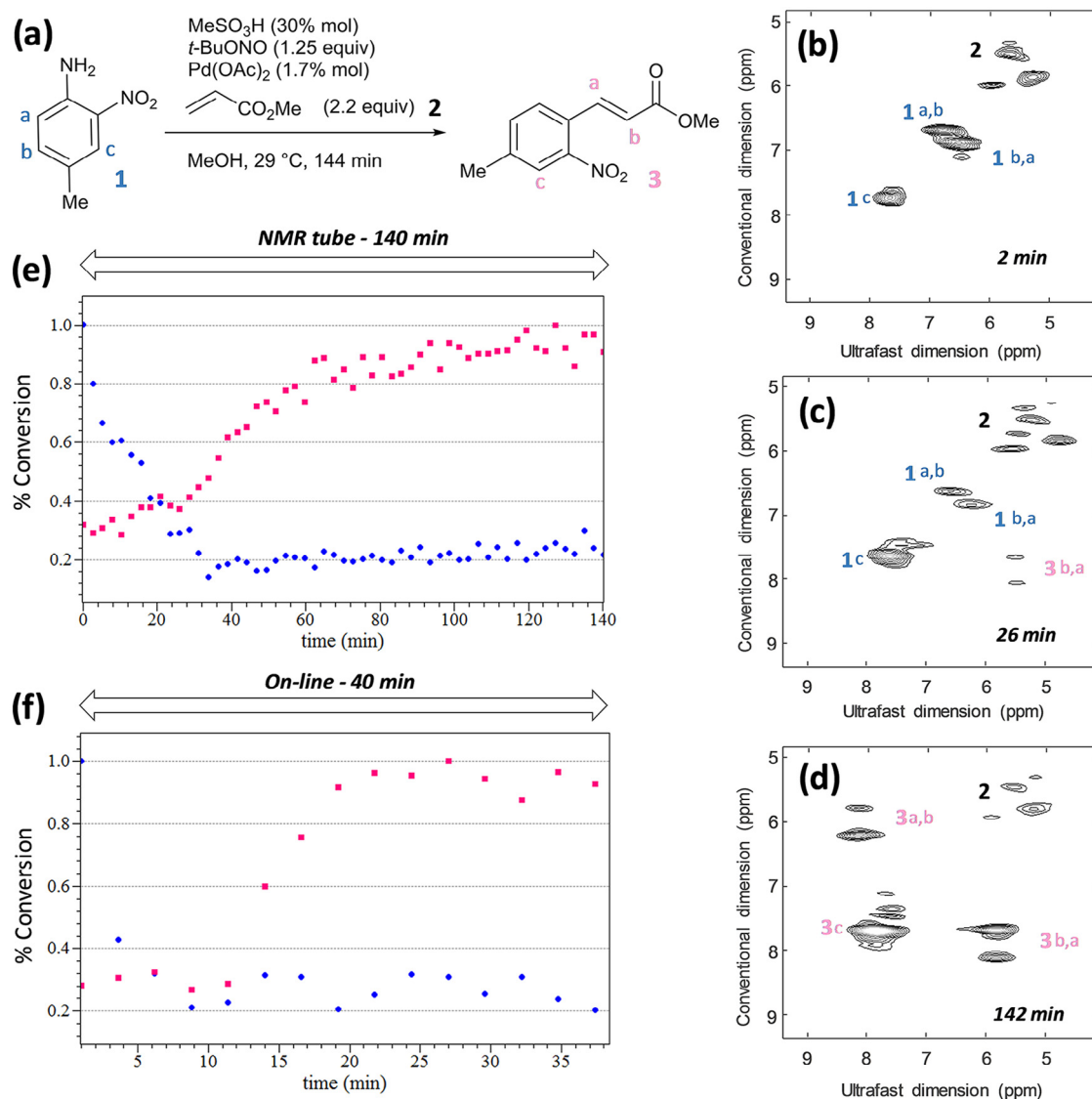


Fig. 5. Scheme of the one pot Heck-Matsuda reaction in the multicomponent mode (a). The reaction was followed at 29 °C for 144 min with an initial concentration of 0.36 mol/L for the limiting starting material 1. The reaction was followed either in the NMR tube or on-flow with mechanical stirring. (b, c, d) show UF COSY spectra recorded with 36 scans in 2.6 minutes at different times (2 min, 26 min and 142 min, respectively) in the course of the in tube monitoring. For the in tube and on-flow monitoring, 55 and 15 successive spectra were respectively recorded every 2.6 min. (e, f) show the evolution of normalized 2D peak volumes as the reaction progresses, for the “in tube” (e) and on-flow settings (f). These volumes were calculated from the cross peaks 1 a,b and 3 b,a indicated in Fig. 2, corresponding to the starting material and the final product, respectively. The blue open circles show the consumption of aniline and the pink open squares show the formation of the final styrene. Figures (a–e) reproduced from (Gouilleux et al., Analyst 2015, 140, 7854–7858) with permission from The Royal Society of Chemistry.

the curves and the rate constants (as determined in Ref [20]), reveal that the coupling reaction is the rate-limiting step of the process with respect to the diazonium formation.

Although this reaction monitoring achieved inside an NMR tube already provides relevant kinetic information, the experimental conditions remain strongly limited: stirring is impossible, the temperature cannot be adjusted and the spectral quality is highly sensitive to potential heterogeneities. On the contrary, an on-flow monitoring through a bypass system allows to monitor the reaction in a vessel with a greater flexibility regarding the experimental conditions: regulated temperature, mechanical stirring, inert atmosphere, etc. We therefore decided to carry out the same reaction, but monitored in an on-flow fashion. In this case, the reaction takes place inside a conventional flask, which is connected to the benchtop spectrometer thanks to a bypass system. A peristaltic pump generates a continuous flow to refresh the sensitive volume of the

magnet with the reaction mixture in the course of the chemical process (Fig. 6).

Yet, this setup raises further questions about the recorded UF spectra, such as the robustness of the spatial encoding performed on a flowing mixture, as reported for online HPLC-UF NMR coupling [43]. Fortunately, in the Spinsolve Benchtop spectrometer, the gradient-axis used for the spatial encoding is perpendicular to the flow-cell, therefore reducing the sensitivity of the UF experiments to the flow. The spins located in a given sample slice can be considered as undergoing the same magnetic field gradient during the whole pulse sequence. Moreover, compared to the “in tube” monitoring, the mixture homogeneity is less critical, even though special care should be given to avoid that precipitates (here, from the diazonium salt) obstruct the capillary of the bypass system. To compare the results from the on-flow experiment to the previously reported “in tube” monitoring, the reaction was performed, while

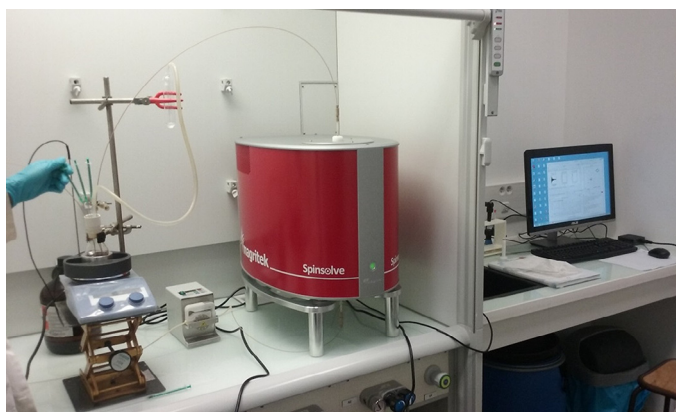


Fig. 6. Picture of the experimental setting for the on-flow monitoring of chemical reactions under the fume hood.

stirring, with identical concentrations (0.36 mol.L^{-1}) and at the same temperature (29°C). The reaction mixture flowed continuously from the flask to the spectrometer at a rate of 2 mL.min^{-1} . This value allows to minimize the time spent by the mixture outside the flask while avoiding some hardware issues such as a significant drift of the carrier frequency which was observed at higher rates. NMR parameters and optimizations – as described in Ref [20] – were kept unchanged, yielding COSY spectra every 2.6 minutes. The kinetic curves obtained with this experimental setup are displayed in Fig. 5f. As for the monitoring carried out inside the tube, the curves highlight the two chemical processes, *i.e* the diazonium formation followed by the Pd-coupling. However, a major difference is the significantly shorter time-scale of the Heck-Matsuda reaction when monitored in an on-flow setting. The plateau is reached at about 40 min, versus 140 min for the “in tube” monitoring. Here, after only 3.6 min, around 60% of the aniline is already consumed and almost 80% of the final styrene is obtained in approximately 17 min. This fast kinetics could be attributed to: i) stirring, which is missing when the reaction takes place in the NMR tube; this result is consistent with recently reported results [44]; ii) the time spent by the

solution in the by-pass system (*ca.* 40% of the reaction time). Inside the tubing, the reaction is carried out in a continuous-flow fashion involving different kinetics than inside the reactor.

5. Screening applications

Although the real-time monitoring of fast chemical processes is probably the most obvious application of UF 2D NMR, another potentially promising utilization of fast 2D experiments at low field relates to screening analysis. Indeed, relevant results about adulteration and authentication issues were recently reported using benchtop spectrometers in different fields, from food matrices – *e.g.* honey [7], edible oils [6], meat [8] – to sexual enhancement and weight loss dietary supplements [9]. In this section we focus on the potential of UF NMR at low field for screening analysis applied to a long-standing concern: the screening of the edible oils. In particular, olive oil, which is a mixture of tri, di, and mono-glycerides, is considered as one of the best sources of fatty acids and natural antioxidants. Its nutritional properties and its excellent taste make olive oil a high-value product that is related to numerous frauds and adulterations [45]. High-resolution ^1H NMR has been widely used in this context for assessing quality and detecting quality adulteration. Moreover, a recent article extended this background to LF NMR [6]. This study emphasized the ability of benchtop NMR to detect the adulteration of olive oils through the addition of cheaper substitutes. 1D spectra recorded in 10 min at 60 MHz and combined with a chemometric approach, enabled to detect hazelnut oil into an olive oil sample at the level of 11.2% w/w. This result encouraged us to investigate the potential of fast 2D experiments to improve the screening performance.

Edible oil is formed of monounsaturated, polyunsaturated and saturated fatty acids, mainly in the form of esters with glycerol (triglycerides), which constitute more than 98% of the total content in the case of olive oil. The assignment of these components is significantly complicated by the overlapped resonances commonly encountered at 43 MHz (Fig. 7). Hopefully, numerous fully assigned high-field ^1H spectra of edible oils are reported in the literature, which facilitates the identification of major peaks [46,47]. An assignment of the visible signals is proposed in Fig. 7 – which

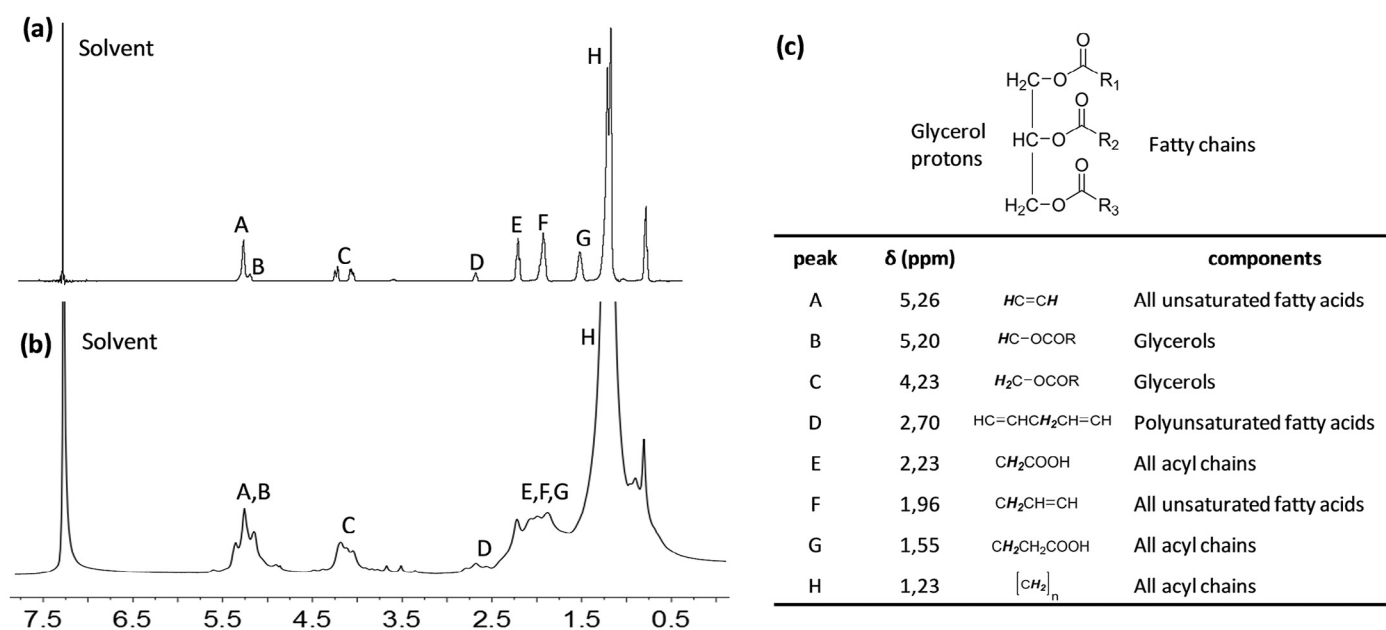


Fig. 7. 1D ^1H NMR spectra of a peanut oil sample (60% in chloroform v/v) recorded at 400 MHz (a) and 43 MHz (b). (c) shows the generic structure of triglycerides, which form the major constituents of edible oils, together with the assignment of main signals based on the literature.

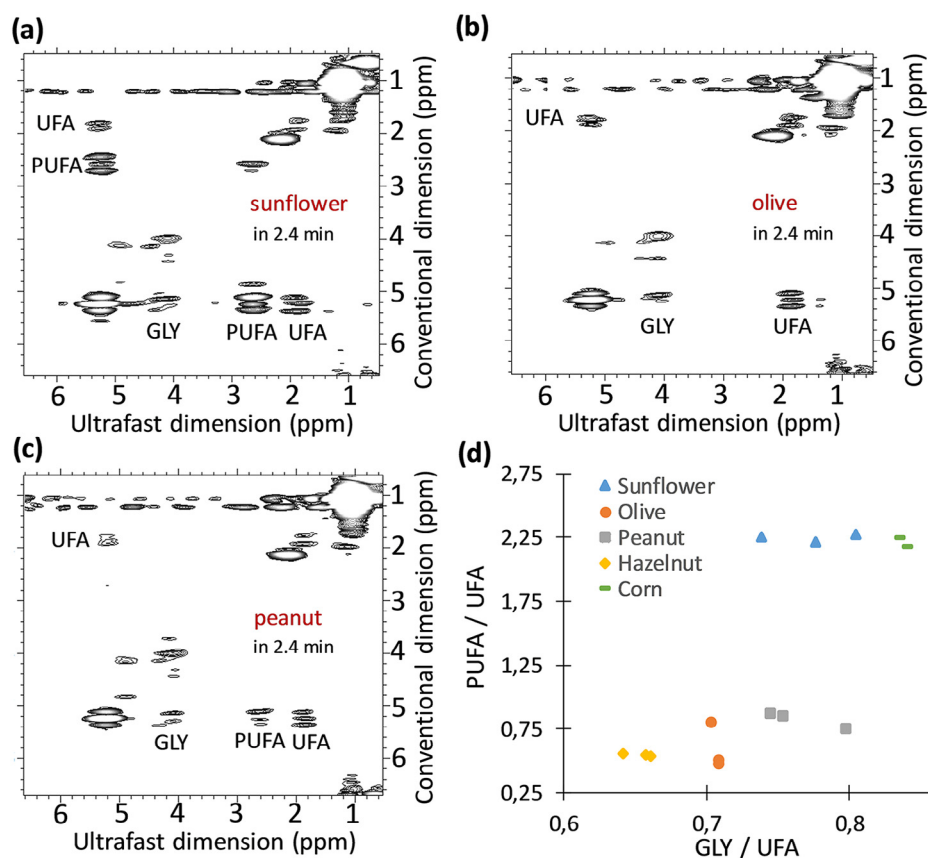


Fig. 8. 2D UF COSY spectra of sunflower (a), olive (b) and peanut (c) oil samples (60% in chloroform v/v) recorded in 2.4 minutes at 43 MHz with the pulse sequence of Fig. 2. (d) shows the PUFA/UFA versus GLY/UFA 2D peak ratios, enabling the group separation of 14 edible oil samples from different origins. UFA and PUFA were calculated by the sum of their symmetric resonances; for GLY only the peak located under the diagonal was taken into account.

compares the LF and HF spectra of the same sample— where some of them are particularly interesting for the present study. The olefinic peak A at 5.26 ppm, arising from protons involved in a double bond, refers to the total number of unsaturated bounds, regardless of whether these are provided by mono- or polyunsaturated chains. The two peaks at 5.20 and 4.23 ppm respectively match with the $-\text{CH}-$ and $-\text{CH}_2-$ groups of the glycerol backbones in the triglycerides. Finally, the peak D located at 2.70 ppm arises from the diallylic CH_2 protons of the polyunsaturated chains (see Fig. 7c).

A convenient way for screening the edible oils at low field is to consider ratios between integrated peaks that reflect the relative concentration of unsaturated fatty acids (noted UFA), polyunsaturated fatty acids (PUFA) and glycerol (GLY) in the oil sample. In practice, a relevant ratio is obtained by dividing the signal C at 4.23 ppm arising from the $-\text{CH}_2-$ protons GLY by the signal A at 5.26 ppm referring to UFA. Although the authors of Ref. [6] used this criterion to screen different edible oils through 1D spectra, this procedure at low field suffers from a measurement bias, since the resonance at 5.20 ppm (peak B) matching with the CH group of the glycerol is totally collapsed with the olefinic peak A (Fig. 7b). In this example, 2D experiments constitute an attractive way to circumvent this overlapping issue, and UF spectroscopy potentially provides a convenient way for such rapid screening. The UF 2D spectra shown in Fig. 8, recorded on different oils in 2.4 min, highlight how the UF COSY experiments overcome the overlapping issue by delivering very well resolved correlation peaks. It is therefore possible to properly determine peak ratios without measurement bias in a short experimental time. It is even possible to discern different oils by a simple visual inspection in the case of peanut,

sunflower and olive oils (Fig. 8), as the intensity of the PUFA cross-peaks increases from the olive to the sunflower oil. We evaluated this promising benefit via a more quantitative approach based on the ratios of integrated peaks. UF COSY experiments were performed on a panel of 14 oil samples from 5 different sources: olive, peanut, corn, sunflower and hazelnut. 36 scans, i.e. 2.4 min, were enough to obtain a SNR higher than 20 for the peaks of interest. J-modulation was tuned to reach a compromise in sensitivity for the cross-peaks of interest arising from UFA, PUFA and GLY. The screening was first based on the ratio GLY / UFA previously reported, and a complementary ratio PUFA / UFA was also determined, reflecting the proportion of polyunsaturated acids in the sample. This ratio could not be accurately determined in 1D because of peak overlaps with the acyl chain signals. A series of 5 UF COSY spectra was acquired for each sample and average peak volume ratios were calculated. The two ratios were plotted in a 2D graph, showing the relevance of this approach to discriminate different sources of edible oils (Fig. 8d). The complementarity of the two ratios is clearly visible in the 2D plot, where the group of olive oils appears well separated from the groups of corn and sunflower oils based on the PUFA / UFA criteria, while the ratio GLY / UFA separates olive from peanut and hazelnut oils.

Naturally, this study does not allow to accurately assess the intra-group variability due to the limited number of samples. Yet, this positive result highlights the relevancy of developing UF 2D NMR on a benchtop spectrometer for screening applications. This urges to investigate, in the near future, further developments along this direction, such as combining UF NMR at low field with chemometrics to build an efficient and rapid screening tool.

6. Conclusion

This article presents some of the main features and application perspectives of ultrafast 2D NMR and its hybrid variant at low field. The excellent analytical performance that can be achieved in a few tens of seconds, and the promising results obtained in the fields of on-flow reaction monitoring and rapid screening, show that this approach can be a powerful complement to existing analytical methods. UF 2D spectroscopy should therefore help increasing the general applicability of benchtop spectroscopy in a variety of situations where the sample complexity generates spectral overlaps – an omnipresent case at low field.

More generally, these first spatially-encoded NMR experiments on a benchtop spectrometer could open the way to a number of developments for a fast analysis of liquid-state samples, including other homo- or heteronuclear UF pulse sequences, pure-shift experiments or diffusion-based measurements. In particular, heteronuclear pulse sequences would give access to a better separation of resonances, albeit at a sensitivity cost. In view of these perspectives, implementing a gradient coil on benchtop spectrometers is likely to become a standard option on low-field spectrometers.

7. Material and methods

7.1. Spectrometers

7.1.1. 400 MHz spectrometer

The experiments performed at high field were recorded on a Bruker Avance III 400 spectrometer operated with Topspin 2.1, at a frequency of 400.13 MHz with a 5 mm BBFO + probe equipped with a z-axis gradient coil generating a maximum field gradient of 0.90 T.m⁻¹.

7.1.2. 43 MHz benchtop spectrometer

The experiments performed at low field were recorded on a Spinsolve from Magritek working at a frequency of 43.62 MHz via a permanent magnet based on a Hallbach design [48,49]. This equipment includes a gradient coil along the B₀-axis (i.e. along the transverse plane of the NMR tube) which can generate a maximum field gradient of 0.16 T.m⁻¹ and operated with the Prosipa Software. The spectrometer allows to work without deuterated solvents thanks to a built-in external fluorine lock system.

7.2. Analytical performance

7.2.1. Model samples

Five samples of commercial 3-bromo-ethylpropionate were prepared in acetone-*d*₆ at different concentrations: 0.5, 1, 2, 3 and 4 mol.L⁻¹.

7.2.2. UF 2D NMR experiments

All the UF spectra are displayed with the indirect domain called “ultrafast dimension” since it results from specific ultrafast features, whereas the direct dimension arising from conventional evolution during the detection period is called “conventional dimension”. The acquisition and processing parameters are given below for each experiment.

7.2.2.1. UF COSY single scan at 43 MHz recorded on the ethanol sample. The UF single scan experiment presented in Fig. 2b was recorded in 400 ms with the homemade pulse sequence described in Fig. 2a, with two 30 ms chirp pulses (i.e. TE = 60 ms) swept over a 3.5 kHz range. The amplitude of the encoding gradients G_e was adapted to obtain a frequency dispersion equivalent to the frequency sweep of the pulses, corresponding to 11.6% of the maximum

gradient strength available. A delay t_{mix} of 42 ms was added prior to the spatial encoding step in order to optimize the effects of J-modulation arising from the constant-time nature of the double spin-echo encoding [42]. During the acquisition, 128 pairs of bipolar gradient pulses G_a were applied (85% of maximum strength, 1280.4 μs each). The desired coherence selection was obtained as described in Fig. 2a. The spectrum was processed using a home-written routine in Prosipa, including an optimized Gaussian apodization in the spatially-encoded dimension [50] and a sine-bell apodization in the FT dimension (see 7.5 for details).

7.2.2.2. Fast hybrid COSY at 43 MHz recorded on the model samples. The UF experiment presented in Fig. 3a was recorded in 12 scans, i.e. in 48 s, with the homemade pulse sequence described in Fig. 2a, with two 40 ms chirp pulses (i.e. TE = 80 ms) swept over a 3.5 kHz range. The amplitude of the encoding gradients G_e was adapted to obtain a frequency dispersion equivalent to the frequency sweep of the pulses, corresponding to 11.6% of the maximum gradient strength available. A delay t_{mix} of 27 ms was added prior to the spatial encoding step in order to optimize the effects of J-modulation. Processing were performed as previously mentioned.

7.2.2.3. UF COSY at 400 MHz recorded on the model samples. The UF experiment presented in Fig. 3c was recorded in a single scan, i.e. in 0.2 s, with the homemade pulse sequence described in Fig. 2a, with two 15 ms smoothed chirp pulses (i.e. TE = 30 ms) swept over a 3.5 kHz range. The amplitude of the encoding gradients G_e was adapted to obtain a frequency dispersion equivalent to the frequency sweep of the pulses, corresponding to 11.6% of the maximum gradient strength available. A delay t_{mix} of 77 ms was added prior to the spatial encoding step so that the magnetizations spend the same time in the transverse plane as in the LF experiments. The spectrum was processed using a home-written routine in Topspin, including an optimized Gaussian apodization in the spatially-encoded dimension and a sine-bell apodization in the FT dimension.

7.2.3. Conventional (Conv) 2D NMR experiments

7.2.3.1. Conv COSY at 43 MHz. The UF experiment presented in Fig. 3c was recorded in 10 min via the commercial routine “QuickCOSY” pulse sequence proposed within the Spinsolve software. The indirect domain was sampled through 256 t_i increments (1 scan each), with each FID containing 8192 points separated by a dwell time of 200 μs. The processing was then automatically performed in MNOVA (Mestrelab Research), including sinusoidal apodization in the two dimensions, followed by a 2D FT to obtain the final spectrum.

7.2.3.2. Conv COSY at 400 MHz. The UF experiment presented in Fig. 3c was recorded in 10 min via the commercial routine pulse-sequence “cosygpqf”. The indirect domain was sampled through 256 increments (1 scan each), with each FID containing 2400 points separated by a dwell-time of 208 μs. The processing was performed under Topspin 2.1, including sinusoidal apodization in the two dimensions, followed by a 2D FT to obtain the final spectrum.

7.3. Reaction monitoring

7.3.1. Procedures for the Heck-Matsuda reaction

7.3.1.1. Monitoring performed inside the NMR tube. *t*-BuONO (0.317 mmol, 38 μL) was added to a solution of aniline (0.252 mmol, 38.8 mg) in MeOH (450 μL). The resulting mixture was stirred and transferred into an NMR tube. Methyl acrylate (0.555 mmol, 50 μL) and palladium acetate (1.7 mol%, 1.1 mg) in MeOH (200 μL) were added to the solution. The catalytic reaction was then initiated by adding methanesulfonic acid (30 mol%, 5 μL). The NMR tube, containing a total volume of 700 μL, was stirred and inserted inside the benchtop NMR at 29°C for 144 min.

7.3.1.2. On-line monitoring procedure. The starting aniline (3.60 mmol, 547.4 mg) was dissolved in MeOH (8.0 mL) inside a 50 mL round-bottom flask. The mixture was stirred for at least 30 min until a homogeneous solution was obtained. The mixture was flowed by the peristaltic pump at a rate of 2 mL/min through the by-pass system consisting of an NMR flow-cell and PEEK tubing for a total volume of 3.8 mL. Then, *t*-BuONO (4.50 mmol, 540 μ L), methyl acrylate (7.92 mmol, 713 μ L) and palladium acetate (1.7 mol%, 13.5 mg) in MeOH (700 μ L) were successively added to the reaction mixture. The bicatalytic reaction was then initiated by adding methanesulfonic acid (30 mol%, 70 μ L). The final 10 mL volume of reactive mixture was regulated at 29°C while stirring for about 40 min.

7.3.2. UF COSY experiments

Fast hybrid UF COSY at 43 MHz. The UF COSY spectra for both reaction monitorings (Fig. 6) were recorded in 146 s (i.e. 36 scans) with the homemade pulse sequence of Fig. 2a with the following features: the spatial encoding was performed by combining two 45 ms chirp pulses (i.e. TE = 90 ms) swept over a 3.5 kHz range with encoding gradients Ge fixed at 9.00% of the maximum gradient strength. The detection was then achieved with 128 pairs of bipolar gradient pulses Ga at 85% of the maximum strength (1280 μ s each). The desired coherence-selection was obtained thanks to both a standard four step phase cycling and gradient pulses. The spectrum was processed in the same way as described for the LF experiments (7.2.2).

7.4. Screening applications

7.4.1. Oil samples

14 samples of edible oils from 5 different sources: olive, hazelnut, sunflower, peanut and corn were purchased from various supermarkets. For each sample, 600 μ L of oil was dissolved in 400 μ L of non-deuterated chloroform inside a 5 mm NMR tube.

7.4.2. UF COSY experiments

Fast hybrid UF COSY at 43 MHz. The UF COSY spectra (Fig. 8) were recorded in 146 s (i.e. 36 scans) with the homemade pulse sequence of Fig. 2a with the following features: the spatial encoding was performed by combining two 30 ms chirp pulses (i.e. TE = 60 ms) swept over a 3.5 kHz range with encoding gradients Ge fixed at 11.6% of the maximum gradient strength. The detection was then achieved with 128 pairs of bipolar gradient pulses Ga at 85% of the maximum strength (1280 μ s each). The desired coherence-selection was obtained thanks to both a standard four step phase cycling and gradient pulses. The spectrum was processed in the same way as previously described for the LF experiments.

7.5. Details of data processing for ultrafast data

All the data recorded in an ultrafast fashion were processed thanks to a home-written routine implemented in Topspin and Prospa. The first step of this routine consists in splitting and rearranging the echoes detected during the successive positive/negative acquisition gradients, leading to two mirror-image $S(k, t_2)$ maps. A sine-bell apodization in the t_2 dimension is applied, followed by a FT (in magnitude mode) yielding two symmetric $S(k, F_2)$ maps. These maps are then added after reversing one of them along the ultrafast dimension (i.e. along k), leading to a 2D spectrum. Finally, a Gaussian apodization in the spatially encoded domain [50] is applied.

The integration of 2D peak volumes was performed by defining rectangular integration regions centered on the maximum of the signal of interest, and optimized to integrate the entire peak of interest while avoiding overlap with other integration regions.

Acknowledgements

The authors warmly acknowledge Magritek, in particular Ernesto Danieli, Juan Perlo and Andrew Coy, for their strong support. We also thank Michel Giraudeau for linguistic assistance. This project was funded by Région Pays de la Loire (Grant RésoNantes).

References

- [1] S.K. Kuster, E. Danieli, B. Blumich, F. Casanova, High-resolution NMR spectroscopy under the fume hood, *Phys. Chem. Chem. Phys.* 13 (2011) 13172–13176.
- [2] V. Sans, L. Porwol, V. Dragone, L. Cronin, A self optimizing synthetic organic reactor system using real-time in-line NMR spectroscopy, *Chem. Sci.* 6 (2015) 1258–1264.
- [3] M.V. Silva Elipse, R.R. Milburn, Monitoring chemical reactions by low-field benchtop NMR at 45 MHz: pros and cons, *Magn. Reson. Chem.* (2015) doi:10.1002/mrc.4189 n/a-n/a.
- [4] F. Dalitz, M. Cudaj, M. Maiwald, G. Guthausen, Process and reaction monitoring by low-field NMR spectroscopy, *Prog. Nucl. Magn. Reson. Spectrosc.* 60 (2012) 52–70.
- [5] E. Danieli, J. Perlo, A.L.L. Duchateau, G.K.M. Verzijl, V.M. Litvinov, B. Blümich, et al., On-line monitoring of chemical reactions by using bench-top nuclear magnetic resonance spectroscopy, *Chemphyschem* 15 (2014) 3060–3066.
- [6] T. Parker, E. Limer, A.D. Watson, M. Defernez, D. Williamson, E.K. Kemsley, 60 MHz 1H NMR spectroscopy for the analysis of edible oils, *Trends Analyt. Chem.* 57 (2014) 147–158.
- [7] R.O.R. Ribeiro, E.T. Mársico, C.S. Carneiro, M.L.G. Monteiro, C.C. Júnior, E.F. Oliveira de Jesus, Detection of honey adulteration of high fructose corn syrup by Low Field Nuclear Magnetic Resonance (LF 1H NMR), *J. Food Eng.* 135 (2014) 39–43.
- [8] W. Jakes, A. Gerdova, M. Defernez, A.D. Watson, C. McCallum, E. Limer, et al., Authentication of beef versus horse meat using 60 MHz 1H NMR spectroscopy, *Food Chem.* 175 (2015) 1–9.
- [9] G. Pagès, A. Gerdova, D. Williamson, V. Gilard, R. Martino, M. Malet-Martino, Evaluation of a Benchtop Cryogen-Free Low-Field 1H NMR Spectrometer for the analysis of sexual enhancement and weight loss dietary supplements adulterated with pharmaceutical substances, *Anal. Chem.* 86 (2014) 11897–11904.
- [10] J. Jeener, Lecture presented at Ampere International Summer School II, Basko Polje, Yugoslavia, DOI (1971).
- [11] W.P. Aue, E. Bartholdi, R.R. Ernst, Two-dimensional spectroscopy. application to nuclear magnetic resonance, *J. Chem. Phys.* 64 (1976) 2229–2246.
- [12] R. Boisseau, U. Bussy, P. Giraudeau, M. Boujtitia, In Situ Ultrafast 2D NMR spectroelectrochemistry for real-time monitoring of redox reactions, *Anal. Chem.* 87 (2015) 372–375.
- [13] L. Frydman, T. Scherf, A. Lupulescu, The acquisition of multidimensional NMR spectra within a single scan, *Prod. Natl. Acad. Sci. U.S.A.* 99 (2002) 15858–15862.
- [14] A. Herrera, E. Fernández-Valle, R. Martínez-Álvarez, D. Molero-Vílchez, Z.D. Pardo-Botero, E. Sáez-Barajas, Monitoring organic reactions by UF-NMR spectroscopy, *Magn. Reson. Chem.* 53 (2015) 952–970.
- [15] L.H.K. Queiroz, D.P.K. Queiroz, L. Dhooche, A.G. Ferreira, P. Giraudeau, Real-time separation of natural products by ultrafast 2D NMR coupled to on-line HPLC, *Analyst* 137 (2012) 2357–2361.
- [16] J.-N. Dumez, J. Milani, B. Vuichoud, A. Bornet, J. Lalande-Martin, I. Tea, et al., Hyperpolarized NMR of plant and cancer cell extracts at natural abundance, *Analyst* 140 (2015) 5860–5863.
- [17] A. Le Guennec, I. Tea, I. Antheaume, E. Martineau, B. Charrier, M. Pathan, et al., Fast determination of absolute metabolite concentrations by spatially-encoded 2D NMR: application to breast cancer cell extracts, *Anal. Chem.* 84 (2012) 10831–10837.
- [18] T. Jézéquel, C. Deborde, M. Maucourt, V. Zhendre, A. Moing, P. Giraudeau, Absolute quantification of metabolites in tomato fruit extracts by fast 2D NMR, *Metabolomics* 11 (2015) 1231–1242.
- [19] P. Giraudeau, L. Frydman, Ultrafast 2D NMR: an emerging tool in analytical spectroscopy, *Annu. Rev. Anal. Chem.* (Palo Alto Calif.) 7 (2014) 129–161.
- [20] B. Gouilleux, B. Charrier, E. Danieli, J.-N. Dumez, S. Akoka, F.-X. Felpin, et al., Real-time reaction monitoring by ultrafast 2D NMR on a benchtop spectrometer, *Analyst* 140 (2015) 7854–7858.
- [21] P. Peluussy, Adiabatic single scan two-dimensional NMR spectroscopy, *J. Am. Chem. Soc.* 125 (2003) 12345–12350.
- [22] A. Tal, L. Frydman, Single-scan multidimensional magnetic resonance, *Prog. Nucl. Magn. Reson. Spectrosc.* 57 (2010) 241–292.
- [23] S. Akoka, P. Giraudeau, Fast hybrid multi-dimensional NMR methods based on ultrafast 2D NMR, *Magn. Reson. Chem.* 53 (2015) 986–994.
- [24] P. Giraudeau, S. Akoka, Sources of sensitivity losses in ultrafast 2D NMR, *J. Magn. Reson.* 192 (2008) 151–158.
- [25] M. Foroozandeh, R.W. Adams, P. Kiraly, M. Nilsson, G.A. Morris, Measuring couplings in crowded NMR spectra: pure shift NMR with multiplet analysis, *Chem. Commun.* 51 (2015) 15410–15413.
- [26] K. Bayle, M. Julien, G.S. Remaud, S. Akoka, Suppression of radiation damping for high precision quantitative NMR, *J. Magn. Reson.* 259 (2015) 121–125.
- [27] J. Workman, M. Koch, B. Lavine, R. Chrisman, Process analytical chemistry, *Anal. Chem.* 81 (2009) 4623–4643.

- [28] F. Susanne, D.S. Smith, A. Codina, Kinetic understanding using NMR reaction profiling, *Org. Process Res. Dev.* 16 (2012) 61–64.
- [29] I.M. Clegg, C.M. Gordon, D.S. Smith, R. Alzaga, A. Codina, NMR reaction monitoring during the development of an active pharmaceutical ingredient, *Anal. Methods* 4 (2012) 1498–1506.
- [30] T.R. Hoye, B.M. Eklov, T.D. Ryba, M. Voloshin, L.J. Yao, No-D NMR (no-deuterium proton NMR) spectroscopy: a simple yet powerful method for analyzing reaction and reagent solutions, *Org. Lett.* 6 (2004) 953–956.
- [31] M. Maiwald, H.H. Fischer, Y.-K. Kim, K. Albert, H. Hasse, Quantitative high-resolution on-line NMR spectroscopy in reaction and process monitoring, *J. Magn. Reson.* 166 (2004) 135–146.
- [32] M.A. Bernstein, M. Štefinović, C.J. Sleight, Optimising reaction performance in the pharmaceutical industry by monitoring with NMR, *Magn. Reson. Chem.* 45 (2007) 564–571.
- [33] M. Khajeh, M.A. Bernstein, G.A. Morris, A simple flowcell for reaction monitoring by NMR, *Magn. Reson. Chem.* 48 (2010) 516–522.
- [34] D.A. Foley, C.W. Doecke, J.Y. Buser, J.M. Merritt, L. Murphy, M. Kissane, et al., ReactNMR and ReactIR as reaction monitoring and mechanistic elucidation tools: The NCS mediated cascade reaction of α -thioamides to α -thio- β -chloroacrylamides, *J. Org. Chem.* 76 (2011) 9630–9640.
- [35] A.C. Barrios, R.T. Sosa, R. Williamson, A. Conway, R. Shankar, T. Sumpter, et al., Safe and efficient synthetic route to a 2,5-dimethyl-1-aryl-1H-imidazole intermediate, *Org. Process Res. Dev.* 15 (2011) 449–454.
- [36] K. Kikukawa, T. Matsuda, Reaction of diazonium salts with transition metals. I. Arylation of olefins with arenediazonium salts catalyzed by zero valent palladium, *Chem. Lett.* 6 (1977) 159–162.
- [37] F.-X. Felpin, K. Miqueu, J.-M. Sotiropoulos, E. Fouquet, O. Ibarguren, J. Laudien, Room-temperature, ligand- and base-free heck reactions of aryl diazonium salts at low palladium loading: sustainable preparation of substituted stilbene derivatives, *Chem. Eur. J.* 16 (2010) 5191–5204.
- [38] F.L. Callonnec, E. Fouquet, F.-X. Felpin, Unprecedented substoichiometric use of hazardous aryl diazonium salts in the heck-matsuda reaction via a double catalytic cycle, *Org. Lett.* 13 (2011) 2646–2649.
- [39] N. Oger, F. Le Callonnec, D. Jacquemin, E. Fouquet, E. Le Grogne, F.-X. Felpin, Heck–matsuda arylation of olefins through a bicatalytic approach: improved procedures and rationalization, *Adv. Synth. Catal.* 356 (2014) 1065–1071.
- [40] N. Oger, E. Le Grogne, F.-X. Felpin, Continuous-flow heck–matsuda reaction: homogeneous versus heterogeneous palladium catalysts, *J. Org. Chem.* 79 (2014) 8255–8262.
- [41] N. Oger, E. Le Grogne, F.-X. Felpin, Handling diazonium salts in flow for organic and material chemistry, *Org. Chem. Front.* 2 (2015) 590–614.
- [42] B. Gouilleux, L. Rouger, B. Charrier, I. Kuprov, S. Akoka, J.-N. Dumez, et al., Understanding J-modulation during spatial encoding for sensitivity-optimized ultrafast NMR spectroscopy, *Chemphyschem* 16 (2015) 3093–3100.
- [43] L.H.K. Queiroz Junior, D.P.K. Queiroz, L. Dhooghe, A.G. Ferreira, P. Giraudeau, Real-time separation of natural products by ultrafast 2D NMR coupled to on-line HPLC, *Analyst* 137 (2012) 2357–2361.
- [44] D.A. Foley, A.L. Dunn, M.T. Zell, Reaction monitoring using online vs tube NMR spectroscopy: seriously different results, *Magn. Reson. Chem.* (2015) doi:10.1002/mrc.4259 n/a–n/a.
- [45] P. Dais, E. Hatzakis, Quality assessment and authentication of virgin olive oil by NMR spectroscopy: a critical review, *Anal. Chim. Acta* 765 (2013) 1–27.
- [46] R. Sacchi, F. Addeo, L. Paolillo, ¹H and ¹³C NMR of virgin olive oil. An overview, *Magn. Reson. Chem.* 35 (1997) S133–S145.
- [47] L. Del Coco, S.A. De Pascali, V. Iacovelli, G. Cesari, F.P. Schena, F.P. Fanizzi, Following the olive oil production chain: 1D and 2D NMR study of olive paste, pomace, and oil, *Eur. J. Lipid. Sci. Tech.* 116 (2014) 1513–1521.
- [48] E. Danieli, J. Perlo, B. Blümich, F. Casanova, Small magnets for portable NMR spectrometers, *Angew. Chem. Int. Ed.* 49 (2010) 4133–4135.
- [49] E. Danieli, J. Perlo, B. Blümich, F. Casanova, Highly stable and finely tuned magnetic fields generated by permanent magnet assemblies, *Phys. Rev. Lett.* 110 (2013) 180801.
- [50] P. Giraudeau, S. Akoka, Sensitivity and lineshape improvement in ultrafast 2D NMR by optimized apodization in the spatially encoded dimension, *Magn. Reson. Chem.* 49 (2011) 307–313.